Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Total heat load inside a control enclosure, the effective surface that can dissipate it for the mounting condition, the internal temperature rise with natural convection and radiation through a stated overall heat-transfer coefficient, the forced-ventilation airflow for a target rise (density corrected for altitude), the fan duty point against a fan curve and the system resistance, optional solar gain, and the cooling-unit capacity needed when the limit cannot be met.

WHAT IT CALCULATES	RELATION
Heat dissipated inside the enclosure, plus any solar gain	$P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$
Effective cooling surface, each face weighted by how freely it sheds heat	$A_{eff} = \Sigma f_{face} \times A_{face}$
Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used)	$\Delta T = P/(h A_{eff})$
Air flow a fan must move to hold the target rise	$V = P/(\rho c_p \Delta T_{target})$, with $\rho = p(z)/(287.055 T)$ and c_p = 1006 J/(kg·K)
Fan duty point, where the fan curve meets the system resistance (found by bisection)	$p_{fan}(V) = p_{sys} (V/V_{ref})^2$
Cooling unit capacity, when the surfaces alone cannot reject the heat	$P_{cool} = P - h A_{eff} (T_{max} - T_{amb})$
Sealed enclosure (sealing temperature T_{seal} entered, no breather), pressure difference across the walls when the air warms or cools from the sealing temperature at constant volume, in absolute temperatures (a rise when $T_{int} > T_{seal}$, a suction when below)	$\Delta p = p_{site} \cdot \lvert T_{int}/T_{seal} - 1 \rvert$
A pressure-compensation element (breather or relief) limits it to its cracking pressure	$dp_{seal} = \min(\Delta p, dp_{relief})$
Force of that pressure on a cover of area A_{cover}	$F_{seal} = dp_{seal} \cdot A_{cover}$

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Effective surface, heat-transfer coefficient and fan-sizing practice (cited, not copied)[2]; Verification of temperature rise[3](10.10).

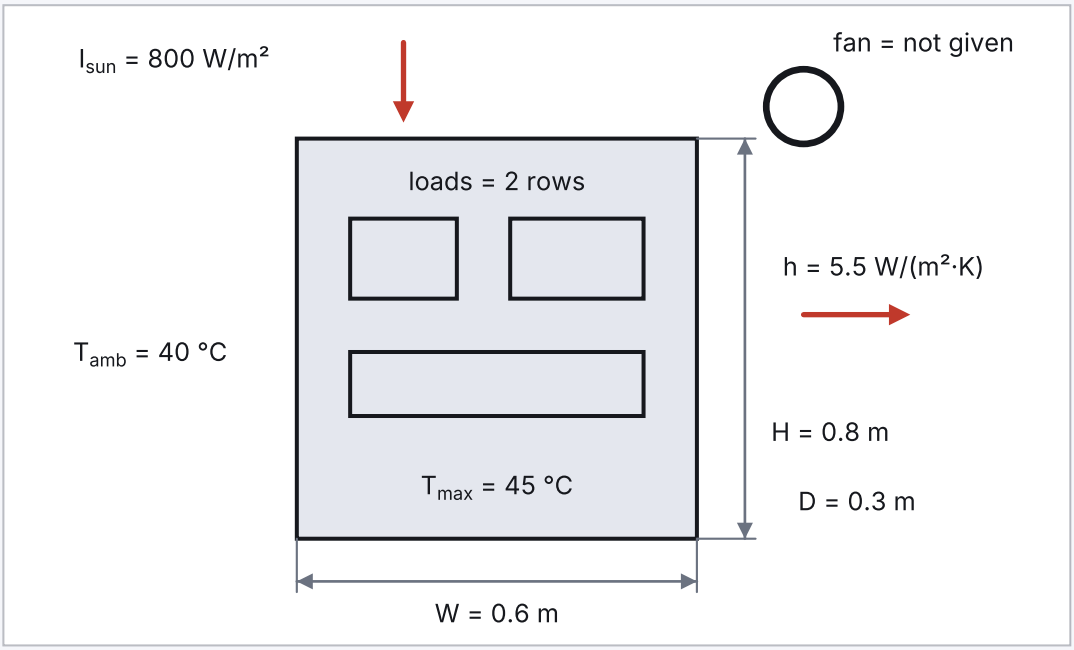


Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-09	any
P_{add}	Additional heat load (linked from E6's dissipated heat or another calculation)	0	W	default	0 … 100000 W
I_{sun}	Solar irradiance on the enclosure (0 = indoors)	800	W/m²	spec: Site conditions SC-5: 800 W/m² on the sunlit side	0 … 1200 W/m²
A_{sun}	Sunlit area	0.48	m²	assumption: One 600 × 800 face in the sun.	0 … 100 m²
α_{sun}	Surface absorptivity	0.5		assumption: Light-grey paint.	0.1 … 1
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
f_{top}	Top exposure (1 free, 0 covered)	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_{front}	Front exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{rear}	Rear exposure (0 = wall-mounted)	1		assumption: Free-standing.	0 … 1
f_{left}	Left side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{right}	Right side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{bottom}	Bottom exposure (0 = on the floor)	0		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
h	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m²·K)	reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K	3 … 8 W/(m²·K)
T_{amb}	Ambient temperature	40	°C	spec: Site conditions SC-5	-20 … 55 °C
T_{max}	Maximum internal temperature	45	°C	spec: Site conditions SC-5	20 … 80 °C
altitude	Site altitude (air density)	0	m	assumption: Free-standing kiosk, bottom on a plinth.	0 … 4000 m
p_{sys}	System resistance (filters, grilles) at the reference airflow	20	Pa	default	0 … 500 Pa
V_{ref}	Reference airflow for the system resistance	100	m³/h	default	1 … 100000 m³/h

TABLE ROW	VALUES
loads[1]	name = PLC and I/O; P = 80 W; qty = 1; duty = 1
loads[2]	name = drives; P = 55 W; qty = 4; duty = 1

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h (natural convection plus radiation to surroundings at the ambient temperature); covered faces dissipate nothing. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h is a reference value the user may edit.)
A2. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise; natural dissipation is neglected for the airflow (conservative). (Manufacturer fan-sizing practice.)
A3. A sealed enclosure (T_{seal} entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed: the pressure difference follows from the ideal-gas relation at constant volume, from the sealing temperature to the internal temperature with natural dissipation T_{int} (the temperature without a

cooling unit); a breather, a seal that leaks or a flexing wall reduces it, so it is an upper bound (a pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance); the site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A4. The fan curve is linear between the entered points; the system resistance scales with airflow squared. (Duty-point idealisation.)
A5. A_{sun} (Sunlit area) is taken as 0.48 m². (One 600 × 800 face in the sun.)
A6. α_{sun} (Surface absorptivity) is taken as 0.5. (Light-grey paint.)
A7. f_{top} (Top exposure (1 free, 0 covered)) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A8. f_{front} (Front exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A9. f_{rear} (Rear exposure (0 = wall-mounted)) is taken as 1. (Free-standing.)
A10. f_{left} (Left side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A11. f_{right} (Right side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A12. f_{bottom} (Bottom exposure (0 = on the floor)) is taken as 0. (Free-standing kiosk, bottom on a plinth.)
A13. altitude (Site altitude (air density)) is taken as 0 m. (Free-standing kiosk, bottom on a plinth.)
A14. The effective surface is the plain exposed-area method of manufacturer practice: each face area weighted by how freely it sheds heat, one overall coefficient h. This is not a verification of temperature rise to IEC TR 60890 or BS EN 61439-1 cl. 10.10 (neither the surface factors nor the standard's temperature-rise correlation is used). (Stated on every run so the result is not read as an assembly verification.)

4. Working

[18] $P_{loads} = \Sigma\ P_i \times n_i \times d_i$
$= 2\ \text{item(s)}$
$= 300\ \text{W}$
Internal heat load

[19] $P_{solar} = \alpha_{sun} * I_{sun} * A_{sun}$
$= 0.5 \times 800\ \text{W/m}^2 \times 0.48\ \text{m}^2$
$= 192\ \text{W}$
Solar gain

[20] $P_{total} = P_{loads} + P_{add} + P_{solar}$
$= 300\ \text{W} + 0\ \text{W} + 192\ \text{W}$
$= 492\ \text{W}$
Total heat to remove

[21] $A_{eff} = f_{top} * W * D + f_{bottom} * W * D + f_{front} * W * H + f_{rear} * W * H + f_{left} * H * D + f_{right} * D$ [1](verification)
$= 1 \times 0.6\ \text{m} \times 0.3\ \text{m} + 0 \times 0.6\ \text{m} \times 0.3\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.8\ \text{m} \times 0.8\ \text{m} \times 0.3\ \text{m}$
$= 1.62\ \text{m}^2$
Effective dissipating surface

[22] $dT_{nat} = P_{total} / (h * A_{eff})$
$= 492\ \text{W} / (5.5\ \text{W/(m}^2\cdot\text{K)} \times 1.62\ \text{m}^2)$
$= 55.219\ \text{K}$
Temperature rise with natural dissipation

[23] $T_{int} = T_{amb} + dT_{nat}$
$= 40\ \text{degC} + 55.219\ \text{K}$
$= 95.219\ \text{degC}$
Internal temperature with natural dissipation

[24] $dT_{allow} = T_{max} - T_{amb}$
$= 45\ \text{degC} - 40\ \text{degC}$
$= 5\ \text{K}$
Allowed rise

[25] u_{nat} = dT_{nat} / dT_{allow}
= 55.219 K / 5 K
= 11.044
Natural dissipation utilisation
[26] $P_{nat,max}$ = h * A_{eff} * dT_{allow}
= 5.5 W/(m²·K) × 1.62 m² × 5 K
= 44.55 W
Heat the enclosure dissipates naturally at the limit
[27] P_{cool} = max(0 W, P_{total} - $P_{nat,max}$)
= max(0 W, 492 W - 44.55 W)
= 447.45 W
Cooling-unit capacity needed if ventilation is not used
[28] p_{site} = 101325 (1 - 2.25577e-5 z)^5.2559
= z = 0 m
= 101.32 kPa
Site pressure
[29] ρ_{air} = p_{site} / (287.055 J/(kg*K) * T_{amb})
= 101.32 kPa / (287.055 J/(kg*K) × 40 degC)
= 1.1272 kg/m³
Air density at ambient and altitude
[30] V_{req} = P_{total} / (ρ_{air} * 1006 J/(kg*K) * dT_{allow})
= 492 W / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K)
= 312.39 m³/h
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] $V_{req,net}$ = max(0 m^3/h, (P_{total} - $P_{nat,max}$) / (ρ_{air} * 1006 J/(kg*K) * dT_{allow}))
= max(0 m^3/h, (492 W - 44.55 W) / (1.1272 kg/m³ × 1006 J/(kg*K) × 5 K))
= 284.11 m³/h
Airflow crediting natural dissipation

5. Results

SYMBOL	VALUE	DESCRIPTION
P_{total}	492 W	Total heat
P_{cool}	447.4 W	Cooling unit needed
$P_{nat,max}$	44.55 W	Natural dissipation at the limit
T_{max}	45 °C	Internal temperature limit
dT_{nat}	55.22 K	Natural temperature rise
T_{int}	95.22 °C	Internal temperature (natural)
V_{req}	312.4 m³/h	Required airflow
P_{loads}	300 W	Internal heat load
P_{solar}	192 W	Solar gain
A_{eff}	1.620 m²	Effective surface
ρ_{air}	1.127 kg/m³	Air density
$V_{req,net}$	284.1 m³/h	Required airflow (net)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural dissipation holds T_{max}	u_{nat} = 11.05	u_{nat} ≤ 1	11.05	

7. Applicability

WARNING T_{max}: natural dissipation holds 44.55 W of 492 W; ventilate or fit a 447.4 W cooling unit. [cooling-unit]

Defaults used (inputs left empty):
P_{add}: not entered, so the default of 0 W is used.
p_{sys}: not entered, so the default of 20 Pa is used.
V_{ref}: not entered, so the default of 100 m³/h is used.

8. Verification

```
Template elec.enclosure-heat@1.5.1 Content hash 6c68c61c7df7f4fe Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. Rittal; nVent Hoffman. Enclosure climate control guides [manufacturer technical literature].
3. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.